Gesture-Controlled Spotify Player

Raspberry Pi Project Proposal

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1. Abstract

This project addresses the limitations of traditional music control systems by developing an inclusive, gesture-controlled Spotify player using a Raspberry Pi 4. Current music platforms primarily depend on touchscreens, physical buttons, or voice commands, which may not be suitable for non-verbal individuals, young children, or users engaged in multitasking activities where hands-free but silent interaction is required. The proposed solution integrates a Raspberry Pi 4 with a camera-based hand-gesture detection system capable of recognising customised movements and translating them into Spotify playback commands such as play, pause, skip, and volume adjustment via the Spotify Web API. To improve usability and user feedback, the system also includes a Live Body Light Visualizer that renders a real-time glowing body-shaped effect on a connected display when a user is present, while gesture control remains active. The expected outcome is an accessible, scalable, and interactive assistive technology solution that promotes inclusive human-computer interaction while demonstrating practical integration of IoT principles, API communication, and real-time computer vision in a Raspberry Pi–based application.

2. Introduction

Digital music streaming platforms such as Spotify have transformed how people access and interact with audio content. Music can now be played instantly across multiple devices, including smartphones, computers, smart speakers, and televisions. However, despite these technological advancements, the primary methods of controlling music playback still rely heavily on touch-based interaction, physical buttons, or voice commands. While effective for many users, these interaction methods are not universally accessible or convenient in all situations. This project addresses the challenge of creating a more natural, accessible, and flexible way to control digital music playback through gesture-based interaction using a Raspberry Pi 4.

The core problem being addressed is the lack of inclusive and adaptable control mechanisms for music streaming platforms. Touchscreen interfaces require physical contact and visual attention, which may not be ideal in situations where users are multitasking, cooking, exercising, or working. Voice assistants provide hands-free control but are not suitable for individuals who are non-verbal, have speech impairments, are in noisy environments, or prefer silent interaction. Additionally, young children may find touch interfaces complex, and voice recognition systems often struggle with accents or unclear pronunciation. These limitations highlight the need for alternative interaction models that do not rely solely on touch or speech.

To complement gesture-based control, the project will provide immediate on-screen feedback through a live body light visualiser. When a person stands in front of the Raspberry Pi camera, the system will detect their body pose and draw an animated, glowing body-shaped

effect (e.g., illuminated skeleton lines and particle highlights) on the monitor. This screen-based approach avoids the complexity and cost of addressable LED hardware while still delivering an engaging “presence-aware” interface that helps users understand when the system is active and which actions have been recognised.

The importance of solving this problem extends beyond convenience. From a social perspective, inclusive design is a key principle in modern computing. Technology should be accessible to people with diverse abilities and communication styles. Creating alternative interaction methods supports accessibility and digital inclusion. From a technical perspective, gesture-based systems demonstrate the integration of embedded systems, sensors, computer vision, and API communication. Economically, gesture-controlled interfaces are increasingly relevant in smart homes, IoT devices, and assistive technologies, making this a valuable area of research and development. The growing popularity of contactless interaction following global health concerns has also increased interest in touch-free systems.

Existing solutions currently available include smartphone apps, smart speakers such as voice-controlled assistants, and wearable devices. Smartphones provide complete playback control through apps, but they require manual interaction and physical handling. Smart speakers allow voice-based commands, offering convenience but introducing privacy concerns and dependency on accurate speech recognition. Some advanced systems use motion detection, such as gaming consoles with camera tracking; however, these are often expensive, designed for entertainment rather than media control, and not easily customisable for everyday music streaming. Additionally, most commercial solutions are closed ecosystems that limit user personalisation.

The main limitations of existing systems include dependence on voice input, limited accessibility for non-verbal users, lack of custom gesture mapping, privacy concerns related to constant audio monitoring, and high cost for advanced motion-tracking devices. Many systems also lack flexibility, preventing users from defining their own preferred interaction patterns. As a result, there is an opportunity to design a low-cost, customisable, and privacy-aware alternative using embedded hardware.

This project proposes the development of a gesture-controlled Spotify player powered by a Raspberry Pi 4. The system will integrate either a gesture sensor module or a camera-based hand tracking solution to detect specific hand movements. These gestures will be processed using Python and relevant sensor or computer vision libraries, then translated into Spotify playback commands through the Spotify Web API or Spotify Connect functionality. Users will be able to personalise gesture-to-action mappings, creating a more adaptable and user-centred experience. In parallel, the camera feed will be used to render a live body light visualiser on a connected screen, providing a glowing body-shaped animation when a user is detected and brief on-screen labels when playback commands are triggered.

By combining embedded hardware, software development, API integration, and human-computer interaction principles, the proposed system aims to deliver a practical and scalable solution for contactless music control. The project not only addresses accessibility challenges but also demonstrates the real-world application of IoT concepts and intelligent input processing. Through this approach, the introduction of gesture-based interaction provides an innovative alternative to traditional music control methods while remaining affordable and technically achievable within a Raspberry Pi–based environment.

3. Related work

Existing projects and implementations demonstrate various ways of using embedded systems such as Raspberry Pi to control media playback, including gesture control and custom interface development. Although these examples illustrate the feasibility of unconventional input methods and streaming integration, they also reveal limitations that the proposed project seeks to address.

One relevant example is a Raspberry Pi–based custom Spotify music player, as illustrated in an online video where the creator configures a Raspberry Pi to stream and control Spotify using a touchscreen interface. The project demonstrates how the Raspberry Pi can be transformed into a dedicated music streaming device that authenticates with Spotify, displays playlists, and controls playback through graphical user interface elements (AKZ Dev, 2025). This implementation confirms the Pi's ability to handle streaming service APIs and user interfaces. However, the interaction model is restricted to touch input, which limits accessibility for users who are non-verbal, prefer contact-free interaction, or are engaged in activities where touch is not desirable.

A further example demonstrates gesture-based media control using an Arduino microcontroller rather than a Raspberry Pi (DIY Tech, 2025). In this implementation, gesture sensors are connected to an Arduino board, which interprets hand movements and sends commands to control media playback. Arduino-based systems are efficient for handling simple sensor input due to their low power consumption and real-time processing capabilities. However, Arduino boards have limited computational resources and lack native operating system support, making them unsuitable for running complex computer vision algorithms or directly integrating with internet-based streaming services such as Spotify. As a result, while Arduino solutions effectively demonstrate gesture detection, they typically require an additional device (such as a computer) to handle higher-level processing and streaming integration. This highlights a key distinction between microcontroller-based systems and single-board computers like the Raspberry Pi.

A related project available on the Instructables platform, Wavepad – Gesture-Controlled Raspberry Pi Music Play, uses sensors attached to a Raspberry Pi to detect hand movements and control music playback (Smith, 2021). This implementation applies hardware sensors (e.g., ultrasonic or proximity sensors) to capture simple gestures and map them to basic actions such as play/pause, skip, or volume control. While this project demonstrates that embedded sensors can be used for gesture interaction, it encounters typical limitations of sensor-based systems: a limited gesture vocabulary, sensitivity to environmental conditions, and minimal capability for recognising more complex or personalised gestures.

When comparing these approaches, distinct differences and limitations become evident. The touchscreen Spotify player shows that embedded systems can integrate with streaming platforms but relies on traditional interaction methods rather than gesture or contact-free input. Camera-based gesture systems display more natural interaction but remain focused on local media and predefined gesture sets, with no mechanism for streaming integration or custom gesture assignments. Sensor-based systems can detect simple movements but offer a narrow range of recognisable gestures and lack adaptability for different users.

These existing implementations underscore key gaps. First, while Raspberry Pi is shown to be a capable platform for both media streaming and gesture recognition independently, few systems combine these into a single, unified solution that supports commercial streaming (such as Spotify) with non-contact gesture input. Second, current gesture systems often use fixed gesture sets without personalisation or mapping flexibility, which limits usability for diverse user groups. Third, there is minimal emphasis in these projects on robustness measures such as false positive reduction, confidence thresholds, and user feedback, which are critical for reliable real-world interactions.

Related research in pose estimation and interactive visual feedback suggests a practical pathway for the proposed visualiser. MediaPipe's Pose Landmarker can detect human body landmarks from images or video and outputs landmark coordinates in both image space and 3D world space, enabling lightweight body-structure visualisations without depth cameras (Google, 2025). On Raspberry Pi, the official camera stack (libcamera/Picamera2) provides continuous frame capture suitable for real-time computer vision pipelines (Raspberry Pi, 2025a; Raspberry Pi, 2025b). For rendering, 2D drawing libraries such as Pygame support efficient drawing of lines and points onto a display surface, which can be adapted for glow and particle effects (Pygame Community, 2025). More advanced silhouette-based effects can be created with background subtraction methods available in OpenCV, although these typically require additional tuning and processing budget (OpenCV, 2024).

The proposed project aims to improve upon these limitations by developing a Raspberry Pi–based system that integrates camera-based gesture recognition with commercial streaming control via the Spotify Web API or Spotify Connect capabilities. Unlike the touchscreen player, it eliminates reliance on physical contact; compared to the Instructables sensor approach, it supports a richer and more expressive gesture vocabulary; and relative to the camera gesture project, it extends functionality to cloud-based media services and introduces user-customisable gesture-to-command mapping. Additionally, the proposed system will include usability enhancements such as confidence scoring, cooldown timers, and visual feedback to improve recognition stability and reduce unintended triggers. By focusing on accessibility and inclusive interaction, this project contributes a more practical and adaptable embedded solution than the current body of hobbyist examples suggests.

Comparison of existing projects and Proposed System

	Raspberry Pi Spotify Player **AKZ Dev (2025)**	**Wavepad (Instructables Sensor)** **Smith (2021)**	**Arduino Gesture System**	**Proposed Project** **Raspberry Project (2026)**

			DIY Tech (2025)	
Hardware Platform	Raspberry Pi	Raspberry Pi	Arduino	Raspberry Pi 4
Interaction Type	Touchscreen	Sensor-based gestures	Sensor-based gestures	Camera-based gesture recognition
Streaming Integration	Spotify	Local playback	Local/basic media control	Spotify Web API/ Spotify Connect
Gesture Complexity	None	Simple directional gestures	Simple sensor gestures	Complex hand gestures (Vision-based)
Personalisation	No	No	No	Yes (custom gesture mapping)
Accessibility Focus	No	No	No	Yes (non-verbal users, multitasking users, kids)
Processing Capability	High	Moderate	Low	High
Robustness Controls	Not emphasised	Minimal	Minimal	Confidence threshold, cooldown, feedback
Scalability	Yes (API)	Limited	Limited	Yes (cloud + embedded integration)

4. Proposed System/Methodology

The proposed system is a Raspberry Pi 4–based gesture-controlled Spotify player that enables users to control music playback using hand movements. The system integrates computer vision–based gesture recognition with Spotify's Web API or Spotify Connect functionality to translate physical gestures into digital playback commands.

The methodology combines embedded hardware, computer vision processing, API communication, and user interface feedback to create an accessible and contact-free music control system.

4.1 System Architecture

The system consists of five main layers:

- Input Layer (Gesture Capture)
- Processing Layer (Gesture Recognition)
- Command Mapping Layer
- Spotify Communication Layer
- Output & Feedback Layer

The Raspberry Pi 4 functions as the central processing unit coordinating all components.

4.2 Components required

Hardware components

- Raspberry Pi 4 Model B (4GB RAM or higher)
- Raspberry Pi Camera Module 3 (CSI connection) and ribbon cable
- Small LCD screen (3-5 inch)
- 32GB or 64GB microSD card (Raspberry Pi OS storage)
- USB-C 5V/3A power supply for Raspberry Pi 4
- Micro-HDMI to HDMI cable (for external display)
- Raspberry Pi case for protection and cable management
- External monitor/TV (available in lab or personal equipment)
- Mini External USB Stereo Speaker

Software Components

- Raspberry Pi OS (Operating System runs different applications)
- Python (Main programming language)
- OpenCV (Image processing)
- Picamera2/libcamera for camera capture
- Pygame for real-time rendering
- Media Pipe and Pose landmarker (Hand tracking and gesture detection)
- Spotify Web API/ Spotify Connect (Playback control)
- JSON (Store gesture mapping)

4.3 Data Processing Method

Step 1: Image Capture: The Raspberry Pi Camera continuously captures video frames.
Step 2: Hand Detection: Media Pipe processes each frame and identifies hand landmarks.
Step 3: Gesture Recognition: The landmark positions are analysed to classify predefined gestures (e.g., fist, open palm, swipe left, swipe right).
Step 4: Command Mapping: Recognised gestures are mapped to Spotify commands using a configurable JSON file.
Step 5: API Communication: The system sends HTTP requests to Spotify's Web API to execute playback commands.
Step 6: Feedback: An LED or on-screen message confirms the action to prevent user confusion.

Communication Method

- **Wi-Fi:** Used for internet access and API communication
- **HTTP Requests:** Used to send playback commands
- **Digital Visualisation:** Screen to show all the user's action

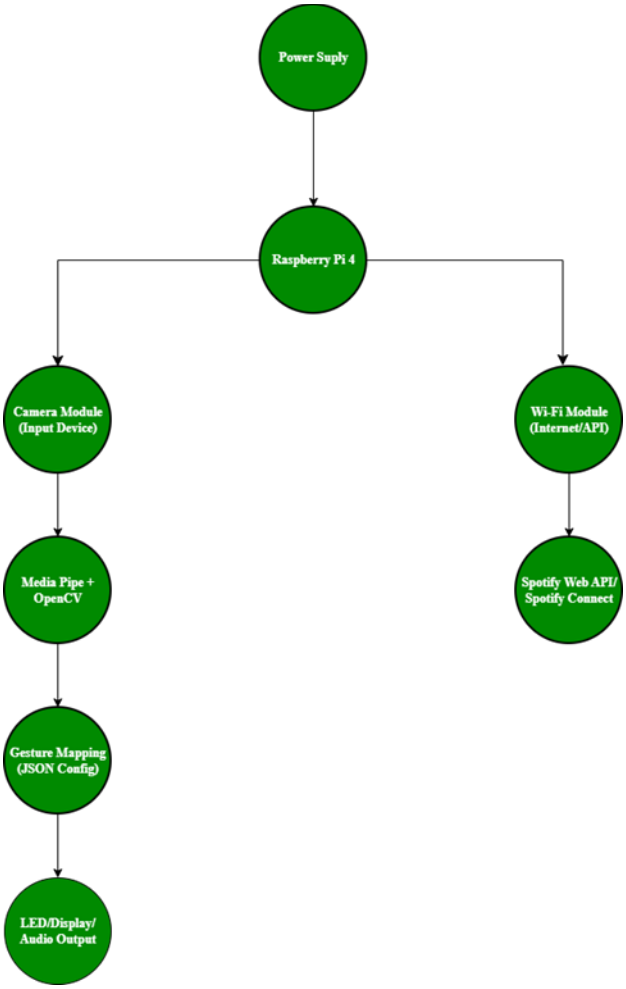
Robustness and Control Logic

- Gesture confidence threshold (Minimum detection score)
- Cooldown timer (e.g. 2 seconds between commands)
- Background noise filtering
- Frame averaging to reduce false positives

Methodology

- Combines embedded systems and cloud APIs
- Uses edge AI processing (on-device gesture recognition)
- Includes robustness mechanisms
- Supports accessibility and personalisation
- Demonstrates IoT and HCI integration

4.4 System diagram



Data Flow Explanation

1. User performs a hand gesture.
2. Camera captures gesture.
3. Media Pipe processes landmarks.
4. Gesture is classified.
5. Mapping layer converts gesture to Spotify command.
6. HTTP request is sent via Wi-Fi.
7. Spotify executes playback action.
8. Feedback is provided via LED/display.

4. 5. Estimate cost for components

Component Name	Qt	Price (€)	Total Cost (€)	Purchase Link
Raspberry Pi Camera Module 3	1	36.89	36.89	https://ie.rs-online.com/web/p/raspberry-pi-cameras/0219259 https://www.sgeducation.ie/raspberry-pi-camera-module-3

3.2" IPS HDMI LCD Display for Raspberry Pi (800x480)	1	32.83	32.83	https://thepihut.com/products/3-2-ips-hdmi-lcd-display-for-raspberry-pi-480x800
Raspberry Pi 4 4gb Starter Kit	-	-	-	(Already available)
Clear Acetate Sheet	1	14.27	14.27	https://www.amazon.ie/dp/B0DYNFP2PQ https://www.amazon.ie/Acetate-Sheets-Colour-Plastic-Reading/dp/B00Y0VCWFG?th=1
Mini External USB Stereo Speaker	1	12.43	12.43	https://thepihut.com/products/mini-external-usb-stereo-speaker
TOTAL			**96.42**	

5. Requirements

5.1 Functional Requirements

- **Gesture Detection:** The system shall detect hand gestures using a camera connected to the Raspberry Pi.
- **Gesture Recognition:** The system shall process captured images using computer vision libraries to recognise predefined hand gestures.
- **Command Mapping:** The system shall translate recognised gestures into music playback commands.
- **Music Playback Control:** The system shall control Spotify playback functions including play, pause, next track, previous track, and volume adjustment.
- **API Communication:** The system shall communicate with the Spotify Web API through an internet connection.
- **User Gesture Customisation:** The system shall allow users to customise gesture-to-command mappings.
- **Visual Feedback:** The system shall provide feedback (e.g., on-screen message) when a gesture is successfully detected.
- **Continuous Monitoring:** The system shall continuously monitor the camera feed to detect gestures in real time.
- **Gesture Filtering:** The system shall ignore unintended movements that do not match predefined gestures.

5.2 Non-functional requirements

- **Performance:** The system should recognise gestures within 1–2 seconds after detection. The system should process video frames in real time to ensure smooth gesture recognition.
- **Reliability:** The system should operate continuously for at least 60 minutes without crashing or interruption. The system should handle temporary gesture recognition errors without stopping the application.
- **Security:** The system should securely authenticate with the Spotify Web API using authorised credentials. Sensitive authentication tokens should not be stored in plain text.
- **Usability:** The system should provide clear feedback to users for at least 1 or 2 seconds when gestures are detected. The system should be simple enough for users with minimal technical knowledge.
- **Scalability:** The system architecture should allow future expansion such as additional gestures, voice commands, or smart home integration.
- **Power Consumption:** The system should run from a standard USB-C 5V power supply without additional power hardware.
- **Maintainability:** The software should be modular so that gesture recognition, API communication, and hardware components can be updated independently.

6. Project Scope

In scope

- Camera-based gesture recognition for a defined set of gestures and also personalized.
- Spotify playback control via the Spotify Web API.
- Live Body Light Visualizer rendered on a connected display.
- Basic configuration for gesture mappings and thresholds.
- On-screen feedback and simple error handling.
- The overall system will demonstrate the integration of embedded hardware, computer vision processing, and cloud-based API communication.

Out of scope (limitations)

- Physical LED strip installations or room lighting control (visuals are screen-based only).
- Multi-user identification or personal profiles.
- Robust operation outdoors or under extreme lighting.
- Full audio-reactive visualisations based on real-time audio analysis.
- Support for music services other than Spotify.
- The system will not implement advanced machine learning models that require extensive training datasets.

Constraints and assumptions

- A stable Wi-Fi connection is required.
- The system will operate primarily on a Raspberry Pi 4 and will rely on a camera module to capture hand gestures.
- Some Spotify Player endpoints require Spotify Premium; testing will be done with a Premium-capable account.
- The camera is positioned in front of the user at a fixed distance (approx. 1–2 metres).
- **Hardware Limitations:** The Raspberry Pi 4 has limited processing power compared to desktop computers, which may restrict the complexity of gesture recognition algorithms that can be implemented.
- The camera-based gesture detection may also be affected by environmental conditions such as lighting or background movement.
- **Software Limitations:** The system relies on external libraries such as OpenCV and Media Pipe for gesture recognition and on the Spotify Web API for playback control.
- **Time Constraints:** The project will be developed within a single academic semester. Due to this limitation, the system will focus on implementing a small set of reliable gestures rather than a large gesture recognition system.
- **Budget Constraints:** The project is designed to be low-cost and accessible. The estimated total hardware cost is approximately €97, which includes the Raspberry Pi 4, camera module, power supply, and other essential components. The system will rely primarily on open-source software libraries to minimise development costs.

6.1 Goals & Objectives

Goal

- Build a low-cost, touchless Spotify controller using a Raspberry Pi and camera-based computer vision for hands-free playback control.
- Provide clear real-time visual feedback through an on-screen “Live Body Light Visualizer” so users can see when they are detected and when commands are recognised.
- Deliver a reliable, modular system that can run in real time on Raspberry Pi hardware and can be extended with additional gestures or visual modes.

Objectives

- Implement camera capture on Raspberry Pi and stream frames to the processing pipeline at ≥ 15 FPS under normal lighting conditions.
- Detect a person in front of the camera using pose landmarks and render a basic glowing skeleton visual effect with detection latency of ≤ 300 ms from frame capture to display update.
- Implement at least 5 gesture commands mapped to Spotify actions: Play/Pause, Next Track, Previous Track, Volume Up/Down (or Mute), each confirmed with an on-screen label.
- Achieve gesture recognition accuracy of ≥ 85% in a controlled indoor test (minimum 50 gesture attempts per gesture across at least 2 users).
- Add a gesture “cooldown” / debouncing mechanism so repeated accidental triggers are limited to ≤ 1 unintended command per 5 minutes during normal use testing.
- Integrate Spotify Web API control and demonstrate successful execution of each command with response confirmation (e.g., playback state change) in ≥ 90% of trials (minimum 30 trials total).
- Demonstrate end-to-end system stability by running the system continuously for at least 60 minutes without crashing, freezing, or losing camera input.
- Document system architecture and evaluation results, including a block diagram, component list, cost table, and a brief test report with measured FPS, latency, and accuracy results.

6.2 Benefits

Who benefits

- Non-Verbal people, kids, adults (Multitasking): can control music without touching a phone/keyboard, which is useful in kitchens, study rooms, offices, living room.
- Users with accessibility needs: gesture control provides an alternative interaction method for users who may struggle with small touch screens, fine motor control, or reaching devices.
- Project stakeholders (lecturer/assessors): demonstrates clear technical learning outcomes (computer vision, real-time processing, API integration, UI feedback, modular architecture).
- Makers / hobbyists / future teams: the design is modular (camera → detection → UI → Spotify API), making it easy to extend for other “touchless control” projects.

Benefits in Use

- Efficiency: faster control of common actions (play/pause/next/volume) without unlocking a phone or navigating apps, especially when the user is moving around the room.
- Safety & hygiene: reduces the need to touch shared devices (mouse/keyboard/tablet) in communal environments; also avoids distraction when the user is cooking, cleaning, or doing tasks.
- Convenience & usability: the Live Body Light Visualizer provides immediate visual confirmation that the camera is detecting a person and that a gesture was recognised (e.g., on-screen label “Next Track”). This reduces confusion and improves user confidence.
- Low cost / high impact: achieved using existing Raspberry Pi hardware + a camera + a standard monitor, without needing physical LED strips or extra electronics. This keeps the build within a semester scope and budget.
- Technical robustness: separating the visualizer layer from Spotify control reduces risk, if the visual effect drops frames, the control module can still function reliably.

What problem it solves effectively

Many music-control setups in shared spaces depend on phones, voice assistants, or touching a device. These approaches can be inconvenient (hands busy), disruptive (voice control in quiet rooms), or unclear (no feedback when detection fails). This project solves that by offering silent, touchless control with clear real-time feedback through the on-screen body-light effect and command prompts.

6.3 Applications of the project

This system is practical anywhere you want touchless, silent control of media with clear visual feedback:

Smart home entertainment

- Living room “gesture remote” for music control on a TV/monitor.
- Party mode: the visualiser acts as a lightweight “human reactive” display while music plays.

Student accommodation / shared spaces

- Dorm common rooms or kitchens where users want quick control without handling phones.
- Study rooms where voice control is not appropriate; silent gestures are less disruptive.

Public or semi-public interactive displays

- Events / exhibitions / museum installations: a person steps in front of the display, and the body-light effect appears, creating an engaging interactive experience while controlling an audio track.
- Retail/demo kiosks: touchless interaction can reduce device wear and improve hygiene.

Fitness / dance / rehearsal spaces

- Users can change tracks during training without touching devices (sweaty hands / gloves), while the visualizer provides an interactive “presence” effect.

Accessibility-focused interfaces

- Proof-of-concept for camera-based input where gesture control can complement or replace small touchscreens or complex menus.

7. Conclusion

The increasing demand for more natural and accessible interaction with digital technologies highlights the limitations of traditional music control methods that rely on touchscreens or voice commands. These interaction methods may not be suitable for all users, particularly non-verbal individuals, children, or people who are multitasking and unable to interact directly with a device. Addressing these limitations requires alternative approaches that enable intuitive and contact-free control of digital media systems.

This project proposes the development of a gesture-controlled Spotify player using a Raspberry Pi 4. The system integrates computer vision technologies with embedded hardware to recognise hand gestures and translate them into music playback commands. By combining Raspberry Pi hardware, camera-based gesture detection, and the Spotify Web API, the system demonstrates how embedded computing and modern software frameworks can be used to create an innovative human–computer interaction solution.

The expected impact of the project is the development of an accessible and user-friendly system that enables users to control music playback through simple hand gestures without physical interaction. This approach can improve accessibility, support inclusive design, and demonstrate the potential of gesture-based interfaces in embedded systems and Internet of Things applications.

The project is technically feasible due to the availability of affordable hardware, open-source computer vision libraries, and well-documented APIs. With careful implementation and testing, the proposed system can be successfully developed within the available time and resource constraints while demonstrating the integration of embedded systems, computer vision, and cloud-based services.

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